

Air-Powered Rockets

Elementary STEAM Field-Trip Packet

Compressed-air tube rockets · Not stomp-rockets

Audience	Mixed grades 2–5
Length	About 3–4 hours (field trip)
Launch system	Air compressor tank + adapter for rocket bodies
Bodies	~10 mm cardboard or plastic tubes
Max pressure	120 PSI (do not exceed)
Launch staffing	1 adult : 5 students
Brand	STEAMSPACE @ Ft Apache

Teacher + student packet — overview, safety, build steps, launch rotations, data sheet, and sources. Site specs confirmed: **~10 mm tubes**, **≤120 PSI**, **1:5** at the launch line.

1. Teacher overview

Students design and build lightweight rockets from ~10 mm tubes, then launch them using stored compressed air. The big science idea: when air pushes one way, the rocket moves the other way (Newton's third law). Fins and careful building help rockets fly straighter. Students test, measure, redesign, and try again — the same habit engineers use.

Learning goals

- Explain rocket motion with action/reaction language (thrust).
- Build a ~10 mm tube rocket with a sealed nose and fins that fit the launch adapter.
- Follow range safety rules; understand why adults control the compressor (≤ 120 PSI).
- Change one design variable and compare flight results (grades 4–5 emphasize fair tests).

How this differs from stomp-rockets

Stomp-rockets use a foot pump / air bladder. This activity uses a compressor tank and adapter. Same family of ideas (air push $\rightarrow$ flight), different hardware and safety controls.

Suggested 3–4 hour arc

Block	Time	Focus
Welcome & hook	15–20 min	What rockets need to fly
Safety briefing	15 min	Range, eyes, adult launch, ≤ 120 PSI, 1:5
Science mini-lesson	20–25 min	Thrust, fins, Newton 3
Build 1	45–60 min	10 mm tube, nose, fins, fit check
Launch rotation 1	40–50 min	First flights + record
Break	15 min	Snack / stretch
Redesign	30–40 min	Change one variable
Launch rotation 2	30–40 min	Compare results
Share-out & clean	20–25 min	What worked? Pack up

2. Safety & launch rules

Site specs (confirmed): ~10 mm tubes · max 120 PSI · 1 adult : 5 students at the launch line

Non-negotiables

- Adults only operate the compressor valve / launch control.
- Clear range — everyone behind the spectator line; nobody downrange during launch.
- Eye protection for launch crew and nearby students (follow site policy).
- Never aim at people — tube rockets can poke eyes (NASA Glenn paper-rocket safety note).
- Inspect before flight — sealed nose, secure fins, snug (not forced) fit on the ~10 mm adapter.
- Pressure ceiling: **120 PSI** — verify gauge before the first student launch; do not exceed.
- Staffing: keep the active launch group at **1 adult : 5 students** (launch adult + spotter preferred).
- One rocket on the pad at a time; recover only when an adult calls the range safe.

Launch line roles

Role	Does
Launch adult	Loads rocket, confirms clear range, fires, announces safe
Spotter adult	Watches spectators / recovery path
Student flyer	Hands rocket to adult; records result on data sheet
Queue captain	Keeps build groups rotating calmly

3. Build guide — tube rockets

Adapted from classroom paper-rocket builds (NASA Glenn Paper Rockets; TeachEngineering Strawkets) for ~10 mm cardboard/plastic tubes and a compressor adapter.

Materials (per student / small team)

- ~10 mm cardboard or plastic tube (must slide onto the adapter smoothly)
- Paper or cardstock for fins and optional nose cone wrap
- Tape (cellophane / masking), scissors
- Soft nose padding (cotton, foam, or clay used carefully — nothing hard/sharp)
- Markers for names and designs
- Clipboard + data sheet (page 6)

Build steps

1. **Fit check first.** Confirm the open end slides onto the adapter smoothly — not floppy, not forced (~10 mm stock).
2. **Seal the nose.** Close the top end so air cannot leak forward (fold + tape, or add a soft cone). Check for leaks.
3. **Add fins.** Tape 3–4 fins near the open (rear) end, evenly spaced, straight back.
4. **Pad the nose.** Soft tip only — protects people and the rocket on landing.
5. **Name it.** Write names so teams reclaim the right rocket.
6. **Adult inspection.** Nose sealed, fins secure, fit OK, no sharp points.

Design variables to try (redesign round)

Change **one** thing at a time: fin size/count, nose mass (soft only), body length (if stock allows), or fin placement. **Do not** use pressure as a student variable — adults keep the system at a set pressure ≤ 120 PSI. Grades 4–5: predict → test → explain.

4. Science talk (tiered)

For everyone (grades 2–5)

Rockets move because of a push. When the launcher releases compressed air, air rushes out and the rocket shoots forward. Scientists call this action and reaction (Newton's third law). The push is called thrust. Fins help the rocket stay steady, like feathers on an arrow.

Stretch for grades 4–5

- Heavier rockets need more push to go as far (connect to force and mass qualitatively).
- Engineers redesign after tests — first try is rarely the final design (TeachEngineering design lessons).
- Fair tests: change one variable; keep launch setup the same when comparing.

Discussion prompts

- What did the air do? What did the rocket do?
- Which fin design flew straighter? Why might that be?
- If we could change only one thing next time, what would you choose?

5. Student data sheet

Name(s): _____ Grade: _____ Date: _____

Rocket name: _____

Flight 1 — first design

Sketch your rocket:

What I changed for Flight 2 (one thing): _____

Flight	Distance / zone	Straight or wobbly?	Notes
1			
2			
Bonus			

Because... (why do you think Flight 2 was different?):

6. Facilitator prep checklist

- Compressor tank inspected; hose/adaptor ready; gauge set and verified ≤ 120 PSI
- Launch range marked; spectator line; recovery path clear of hazards
- Eye protection staged; first-aid kit accessible
- ~10 mm tube stock sorted to adapter; spare fins paper/tape/scissors stations
- Data sheets + pencils clipped; demo rocket built
- Adult roles assigned (launch / spotter / rotations) — plan for **1:5** at launch line
- Weather plan (wind limits for outdoor launches)

7. Sources & credits

This packet adapts public educational materials for a STEAMSPACE compressed-air tube-rocket field trip. It does not reproduce full worksheets from those sites.

- TeachEngineering — Rockets curricular unit (Grades 3–5):
https://www.teachengineering.org/curricular-units/cub_rockets_curricularunit
- TeachEngineering — Strawkets and Thrust:
https://www.teachengineering.org/activities/view/cub_rockets_lesson03_activity1
- TeachEngineering — Using Thrust, Weight & Control:
https://www.teachengineering.org/lessons/view/cub_rockets_lesson03
- TeachEngineering — Learn to Build a Rocket... (design constraints):
https://www.teachengineering.org/lessons/view/cub_rockets_lesson05
- NASA Glenn Research Center — Paper Rockets (Gregory Vogt / Roger Storm):
https://www.grc.nasa.gov/WWW/K-12/TRC/Rockets/paper_rocket.html
- Wikipedia — Water rocket (compressed-air energy / Newton 3 background only):
https://en.wikipedia.org/wiki/Water_rocket

Demo videos (prep / hook — not site PSI specs)

- Creation Crate — STEM Class Ideas: Compressed Air Rocket Launcher:
<https://www.youtube.com/watch?v=JIRZjnjUVhc>
- I Like To Make Stuff — How to Make an Air-Powered Rocket Launcher:
<https://www.youtube.com/watch?v=kOnhzuhZGY8>
- Make: — Weekend Project: Compressed Air Rocket: <https://www.youtube.com/watch?v=eNfFK5uo6D0>
- iCreatables — Air Powered Rocket Launcher - How To Build: <https://www.youtube.com/watch?v=9AKM2UksvSk>
- Teach Beside Me — DIY Cardboard Tube Rocket Launcher for Kids:
<https://www.youtube.com/watch?v=6IEGg9Vyymo>
- BeardedScienceGuy — Paper Rockets for Under Five Dollars (Physics):
<https://www.youtube.com/watch?v=AweP9Rbryhs>
- Full annotated list in vault: [sources/video-links.md](#)

STEAMSPACE brand colors from steamspace.vercel.app site CSS (pine, teal, sun, sand). Packet authored for the Global vault Resource Builder topic air-powered-rockets. Site specs from Bruce (2026-08-27): ~10 mm tubes, ≤120 PSI, 1:5 launch staffing.